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Summary of Expressions

Arithmetic Operators

Arithmetic: Perform basic arithmetical operations like addition and multiplication.

Binary arithmetic operators in C are operators that perform standard mathematical calculations on numeric values and floating-point numbers. They require exactly two operands to function. C provides five core binary arithmetic operators to handle addition, subtraction, multiplication, division, and remainder calculations. Making them essential for performing calculations, processing data, and solving computational problems in C programs.

Arithmetic Operator : Addition +

To summarize one of the simple binary arithmetic operators is the addition operation which is in of itself simple and self-explanatory. You add two values or integers together, The input is the numbers, the output is the sum of the two or more values or integers.

```
/*  Fig.  2.5:   fig02_05.c
    Addition program  */
#include  <stdio.h>

int main()
 {
     int integer1, integer2,  sum;          /*  declaration  */

     printf(  "Enter first integer\n" ) ;      /* prompt  */
     scanf(  "%d",  &integer1  );              /* read an integer */
     printf(  "Enter  second integer\n" );  /* prompt  */
     scanf(  "%d",   &integer2  );             /* read an integer */
     sum =  integer1  +  integer2 ;          /* print sum */

      return  0;             /*  indicate that program ended successfully */
}
```

Arithmetic Operator : Subtraction -

To summarize one of the simple binary arithmetic operators is the subtraction operation which is in of itself simple and self-explanatory. You subtract one value or integer from another, The input is the numbers, the output is the difference of the two or more values or integers.

```
#include <stdio.h>

int main ()
{
    int a = 20,  b = 8;  /* declaration*/
    int difference;        /* declaration*/
    difference = a - b;  // 20 - 8 = 12   /*operation*/
    printf("a - b is %d\n", difference);  /* print difference*/

    return 0;      /* indicate that program ended successfully */
}
```

Arithmetic Operator : Multiplication *

To summarize one of the basic arithmetic operations multiplication is that you basically multiply two operands or integers or values. Using an asterisk as the symbol for the operation. This is considered a binary arithmetic operator. The finished answer will be the product.

```
#include  <stdio.h>

int main () {
    int a = 5, b = 2;            /* declaration*/
    int product = a * b;         /*operation*/
    printf("a * b = %d\n", product);      /* print product*/

return 0;         /*  indicate that program ended successfully */
}
```

Arithmetic Operators : Division /

To summarize one of the basic arithmetic operations division is that you basically divide two integers or values from each other. The first value being the dividend and the second value being the divisor. With the answer becoming the result or commonly known as the quotient of the division operation. This is also a binary arithmetic operation.

```
#include <stdio.h>

int main () {
    int a = 14; b = 7;       /* declaration*/
    int result = a / b;      /*operation*/
    printf("a / b = %d", result);        /* print result*/

return 0;        /* indicate that program ended successfully */
}
```

Relational Operators

Relational operations in C compare two values or expressions to find out how they relate to each other. Relational Operations are often used in statements that have to do with decision-making. These are commonly if-else, if, and while. But also, they can be used to control the flow of the programs with using relational operators for loops. The 6 relational operators are equal to, not equal to, less than, greater than, less than or equal to, and last but not least, greater than or equal to.

Relational operator: Compare two values, while logical operators combine or invert conditions.

Relational Operators..cont.

Four basic operators for comparison of values in C. These are typically called relational operators:

> “greater than”
The greater than operator is a relational operator in C that checks whether the first operand is greater than the second operand or not.

< “less than”

The less than operator is a relational operator in C that checks whether the first operand is lesser than the second operand.

>= “greater than or equal to”

The greater than or equal to the operator is a relational operator in C that checks whether the first operand is greater than or equal to the second operand.

<= “less than or equal to”

The less than or equal to the operator is a relational operator in C that checks whether the first operand is less than or equal to the second operand.

```
#include <stdio.h>

int main()  {
    int a = 15;              /* declaration*/
    int b = 10;              /* declaration*/
    int c = 15;              /* declaration*/

    if (a > b) {             /*operation*/
        printf("a is greater than b\n");      /* print result*/
    }
    if (b < a) {             /*operation*/
        printf("b is less than a\n");         /* print result*/
    }
    if (a >= c)  {           /*operation*/
        printf("a is greater than or equal to c\n");    /* print result*/
    }
    if (b <= a) {            /*operation*/
        printf("b is less than or equal to a\n");       /* print result*/
    }
    return 0;        /*  indicate that program ended successfully */
}
```

Relational Operators...cont.

The C equal to operator (==)

is a relational operator that is used to check whether two given operands are equal or not.

The C not equal (!=)

to operator is another relational operator used for checking whether the two given operands are equal or not.

```
#include  <stdio.h>

int main () {
     int score  =  85;                          /* declaration*/

     if(score  ==  100) {                       /*operation*/
          printf("Perfect score!");             /* print result*/
}
     else {                                     /*operation*/
           printf("Not a perfect score");  /* print result*/
}

return 0;        /*  indicate that program ended successfully */
}
```

Logical Operators

A summary of logical operators are that they are used to create compound expressions. There are three logical operators in C

|| “logical OR” A compound expression formed with || evaluates to 1 (true) if any one of its components is true

&& “logical AND” A compound expression formed with && evaluates to true if all of its components are true.

! "Logical NOT" Reverses the truth value of its operand.

```
#include <stdio.h>

int main()
{
        int a = 20,  b = 40;            /* declaration*/

        if ( a > 10  && b > 10 )        /*operation*/
        {
           printf(“Both values are greater than 10\n”)   /*print result*/
        }
        else                                /*operation*/
        {
           printf(“One or both values are less than or equal to 10\n”);  /*print result*/
        }
         return 0;          /*  indicate that program ended successfully */
}
```

Conclusion

In conclusion, arithmetic, relational, and logical operators are important parts of the C programming language. Arithmetic operators allow programmers to perform calculations such as additional, subtraction, multiplication, and division. Relational operators are used to compare values and help programs make decisions based on different conditions. Logical operators allow multiple conditions to be combined or reversed, making programs more flexible. Overall, understanding these operators is important because they provide the foundation for performing calculation, making decisions, and controlling the flow of a C program.